Physiology of Stomach and Duodenum

Department of Physiology, Microbiology & Genetics

Unit of Medical Physiology

College of Medicine / HMU

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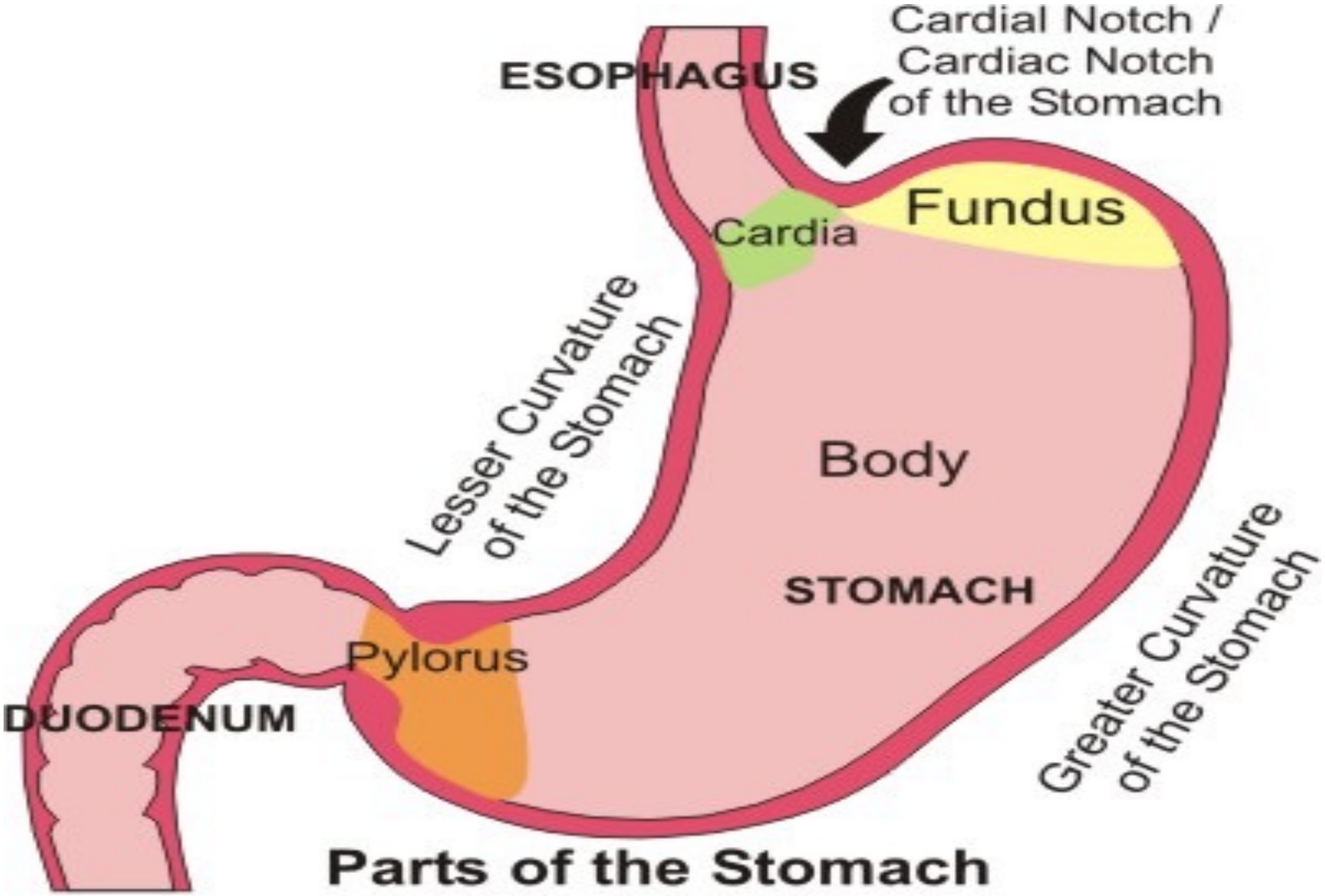
Learning Outcome

- To be able to describe functions of stomach and duodenum.
- Movements in the stomach
- To describe mechanism involved in the control of their functions.

Main Functions of the Stomach

- Bulk storage of undigested food.
- Mechanical breakdown of food.
- Chemical breakdown of food by disruption of chemical bonds via acids and enzymes (pepsin)
- Production of intrinsic factor (aids absorption of vitamin B12).
- Very little absorption of nutrients, some drugs, however, are absorbed
- Enteroendocrine cells that produce hormones (gastrin).



Parts of the Stomach

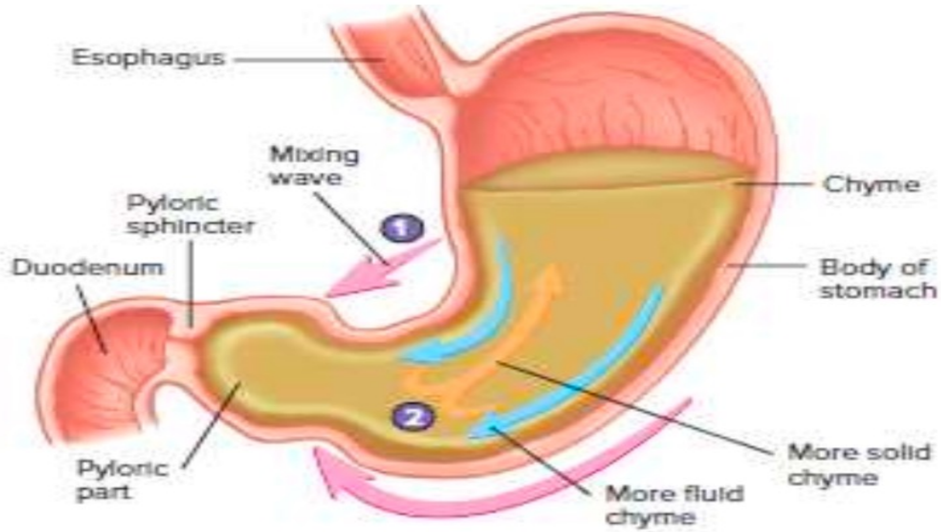
Movements in the stomach

- The stomach consists of fundus, body and pylorus.

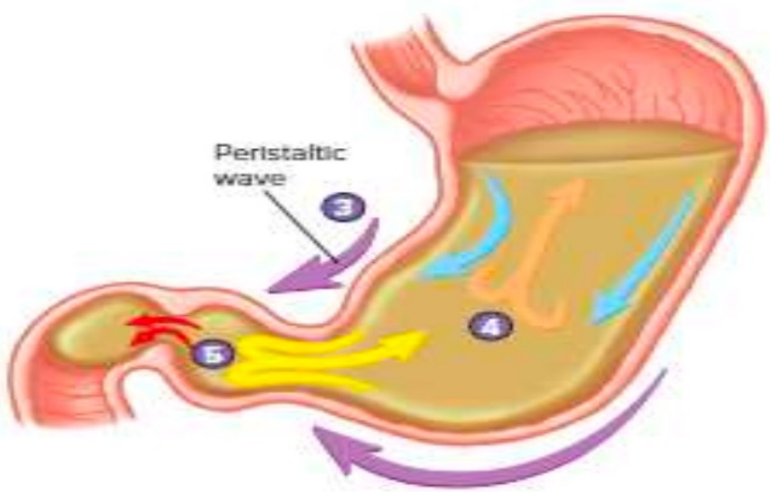
- **Proximal area (fundus and body) has *thin wall*** *and contracts weakly and infrequently* → holds large volumes of food (**to store food**) because of **receptive relaxation**
- **Distal area (pylorus)** has ***thick wall with strong*** and frequent peristaltic contractions that mix and propel food into the duodenum.
- Also, **distal area** is responsible for **gastric emptying** into duodenum.

Movements in the stomach

1. A mixing wave initiated in the body of the stomach progresses toward the pyloric sphincter (*pink arrows directed inward*).
2. The more fluid part of the chyme is pushed toward the pyloric sphincter (*blue arrows*), whereas the more solid center of the chyme squeezes past the peristaltic constriction back toward the body of the stomach (*orange arrow*).

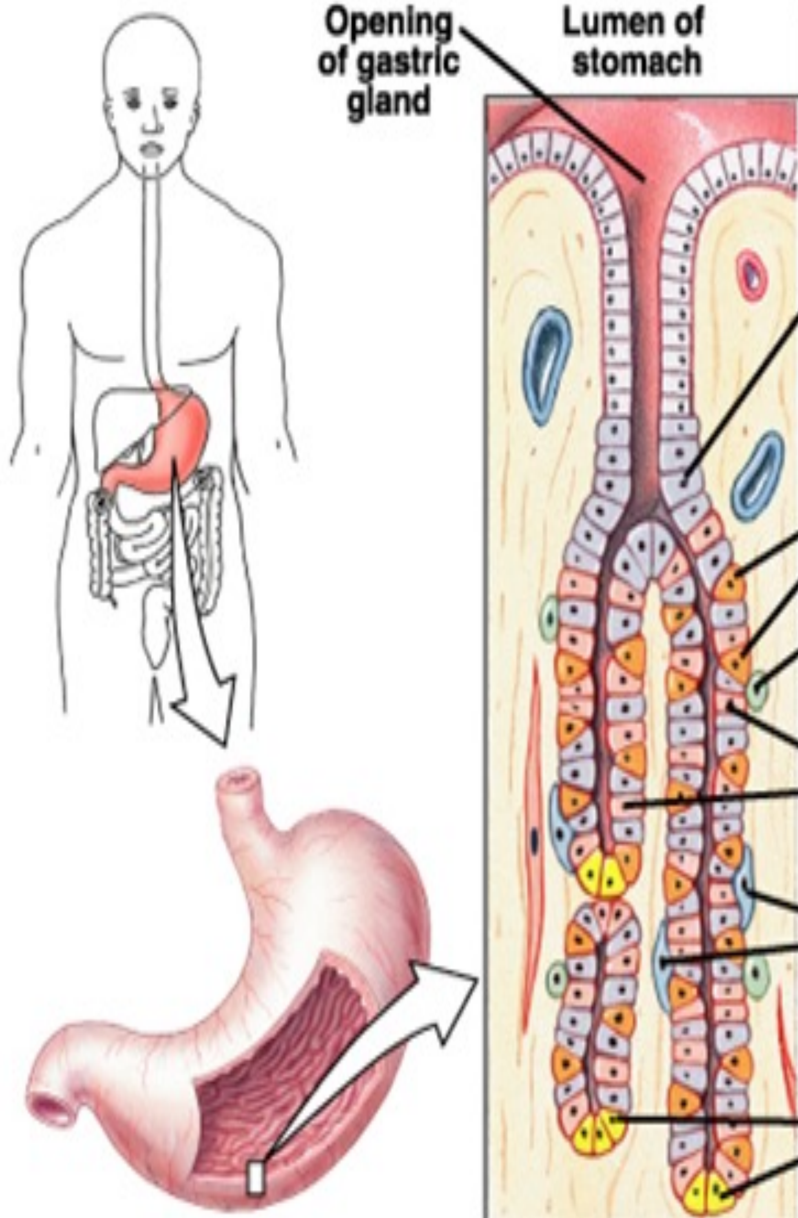


3. Peristaltic waves (*purple arrows*) move in the same direction and in the same way as the mixing waves but are stronger.
4. Again, the more fluid part of the chyme is pushed toward the pyloric region (*blue arrows*), whereas the more solid center of the chyme squeezes past the peristaltic constriction back toward the body of the stomach (*orange arrow*).
5. Peristaltic contractions force a few milliliters of the mostly fluid chyme through the pyloric opening into the duodenum (*small red arrows*). Most of the chyme, including the more solid portion, is forced back toward the body of the stomach for further mixing (*yellow arrow*).



Gastric Secretion

- The total volume of GIT secretions is about 6-8 L/day.
- Secretions arise from specialized cells lining the GI tract, the pancreas, liver and gallbladder.
- GI secretions function to:
  - *Lubricate* (water and mucus),
  - *Protect* (mucus),
  - *Sterilize* (HCl),
  - *Neutralize* (HCO_3^-), and
  - *Digest* (enzymes).



Source	Substance Secreted	Function
Mucous neck cell	Mucus	Physical barrier between lumen and epithelium
	Bicarbonate	Buffers gastric acid to prevent damage to epithelium
Parietal cells	Gastric acid (HCl)	Activates pepsin; kills bacteria
	Intrinsic factor	Complexes with vitamin B_{12} to permit absorption
Enterochromaffin-like cell	Histamine	Stimulates gastric acid secretion
Chief cells	Pepsin(ogen)	Digests proteins
	Gastric lipase	Digests fats
D cells	Somatostatin	Inhibits gastric acid secretion
G cells	Gastrin	Stimulates gastric acid secretion

Gastric Hormones

- Chemical messengers that regulate intestinal and pancreatic function.
- The “gut” is the largest endocrine organ in the body.
- The messengers can act as:
    - *Endocrine*: distant target
    - *Paracrine*: close target
    - *Autocrine*: self target
    - *Neurocrine*: neurotransmitter or stimulator.

Gastric Hormones: conti..

- Stimulus for release is: FOOD

- Composition of food dictates timing and specific hormone release.

Gastric Hormones:

- **Gastrin.**
- **Somatostatin.**
- **Gastrin-releasing peptide (GRP).**
- **Histamine.**

Gastrin:

SYNTHESIS: G-cells in the antrum.

RELEASE (Stimulating Factors):

- AA, protein, vagal tone, antral distention, GRP, pH > 3.0, ETOH (Ethyl Alcohol) , Histamine.

INHIBITION (Inhibitory Factors):

- pH < 3.0, somatostatin, secretin, CCK, VIP, GIP, glucagon.

TARGET CELLS:

- Parietal and Chief cells.

Gastrin: conti..

Action(s):

1. Stimulates acid secretion.
2. Direct action on parietal cells.
3. Potentiating interaction with histamine, possible releasing of histamine.
4. Increases release of electrolytes & water from stomach, pancreas, liver and Brunner’s glands.
5. Stimulates motility in stomach, intestine, and gall bladder.
6. Inhibits contraction of pylorus and sphincter of Oddi.
7. Stimulates GI mucosal growth.

Somatostatin

SYNTHESIS:

- CNS, antrum, fundus, SI, colon, and D-cells in pancreas.

RELEASE:

- Antral acidification.
- Fats, protein, acid in duodenum.
- Pancreatic: glucose, amino acids, CCK.

INHIBITION:

- Release of acetyl-choline from vagal nerve fibers.

Somatostatin: **conti…**

ACTION(S):

1. The "master off switch"
2. Inhibits the release of most GI hormones
3. Inhibits pancreatic and GI secretion(s)
4. Inhibits intestinal motility.

CLINICAL:

- Octreotide- decrease fistula output.
- Treatment of esophageal variceal bleed.
- Can ameliorate symptoms of endocrine tumors.

GRP (Gastrin Releasing Peptide)

SYNTHESIS:

Gastric antrum, small bowel mucosa

RELEASE: vagal stimulation

ACTIONS:

- Stimulate release of all GIT secretions.
- Stimulate GIT motility.

Histamine

STIMULATES parietal cells

Found in the acidic granules of ECL cells and resident Mast cells.

Release is **STIMULATED** by:

Gastrin, acetyl-choline, epinephrine

INHIBITED by Somatostatin.

Mucus secretion

- Soluble and insoluble mucus are secreted by cells of the stomach.
- ***Functions***:
- **Soluble mucus** mixes with the contents of the stomach and helps to lubricate chyme.
- **Insoluble mucus** forms a protective barrier against the high acidity of the stomach content.

Intrinsic Factor: Help absorption of vitamin B12.

Acid Secretion

- Stimulated Acid Secretion: which has three Phases:
    - Cephalic phase
    - Gastric phase
    - Intestinal Phase

Cephalic Phase:

Originates with the sight, smell, thought or taste of food.

- Stimulates the cortex and hypothalamus.
- Signals cause Vagus to release Ach, Ach causes increase in parietal cell acid production.
- Accounts for 20-30% of acid production.

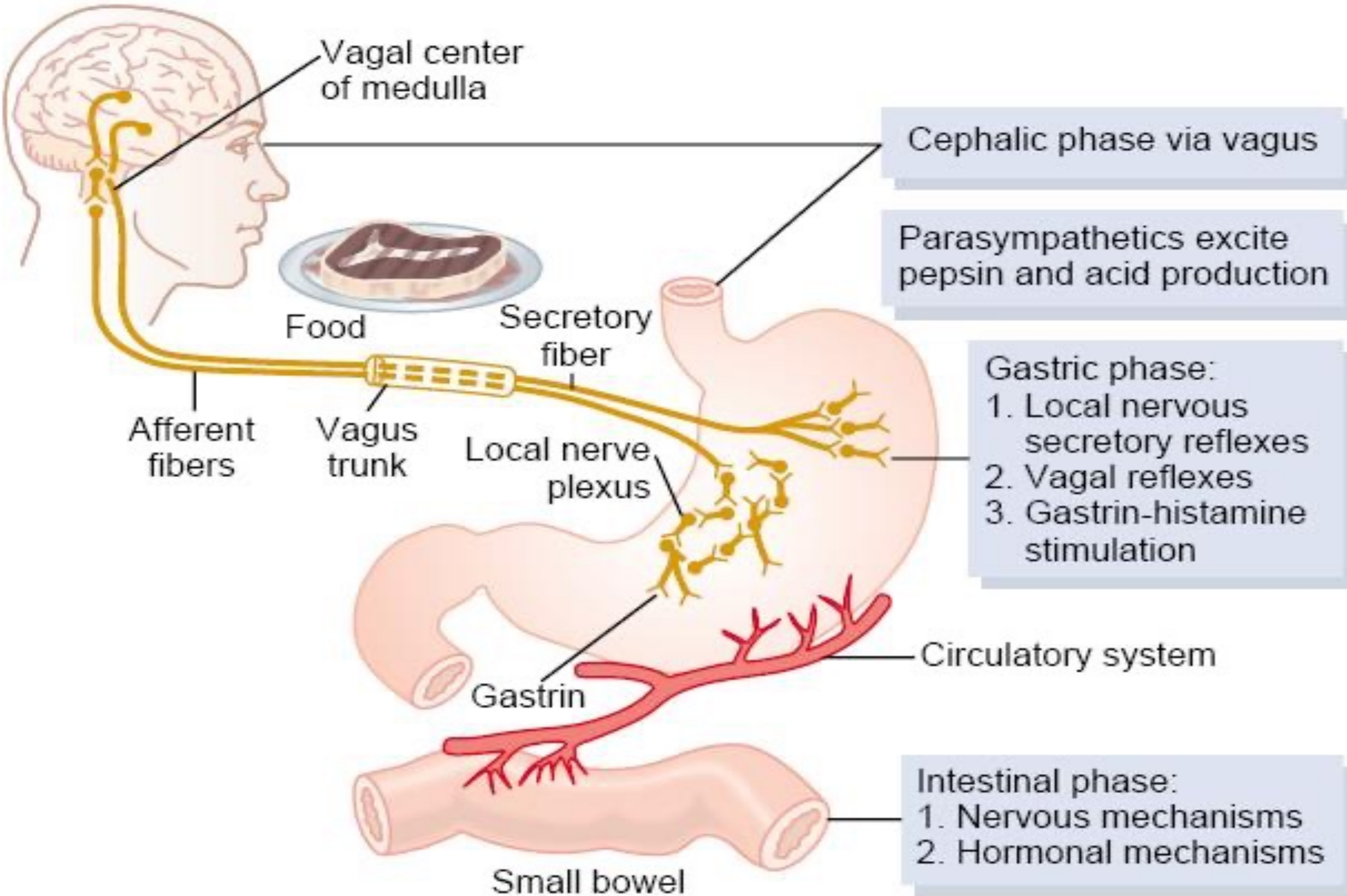
Gastric Phase

- Begins when food enters the gastric lumen (gastric distention).
- Digestion products stimulate the G cells, they release gastrin, parietal cells release acid.
- Distention alone can increase acid production.
- Accounts for 60-70% of acid production.
- Phase lasts until the stomach is empty.

Intestinal Phase

- Poorly understood.
- initiated by chyme entering the small bowel.
- Accounts for ~10% of acid production.

Regulation of Gastric Secretion



Functions of Gastric HCL

1. Activates pepsinogen into pepsins.
2. Provides optimum for pH for action of pepsins.
3. Denatures protein denaturation → help its digestion.
4. Kills bacteria in food.
5. Help Fe^{2+}、Ca^{2+} absorption.
6. Promotes pancreatic, SI.l and bile secretion.

Small intestine Secretion

- Secretion from duodenal gland and intestinal gland.
- Secretory volume is 1～3L/day.
- It contains inorganic ion, mucoprotein, IgA, various enzyme, e.g. enterokinase ,etc
- **Function:**
  1. Protective effect by mucous.
  2. Digestion by enzymes such as peptidase, sucrase, lipase.
  3. Dilution.

Duodenum

- Its main function is to receive the chyme from the stomach and continues the digestion.
- Receives bile drained from the liver and gallbladder and pancreatic juice secreted by the pancreas.
- These secretions aid in digestion of food.
- Regulates the rate of gastric emptying and triggers the hunger signals.
- Both of these functions are performed with the help of hormones that are produced and released by the duodenal epithelium.

Duodenum: conti…

- Secrete two hormones known as *secretin and cholecystokinin.*
- Both hormones encourage secretion of bile and pancreatic juice.
- Absorbs the nutrients and it does it even more than the stomach.
- Because of that, in obese people, the duodenum is frequently bypassed in gastric bypass surgery to decrease the absorption of nutrients.

Summary

1. The stomach connects the esophagus to the duodenum.
2. The principal anatomical regions of the stomach are the cardia, fundus, body, and pylorus.
3. Adaptations of the stomach for digestion include rugae; glands that produce mucus, hydrochloric acid, pepsin, gastric lipase, and intrinsic factor; and a three-layered muscularis.
4. Mechanical digestion consists of mixing waves.
5. Chemical digestion consists mostly of the conversion of proteins into peptides by pepsin.
6. The stomach wall is impermeable to most substances.
7. Among the substances the stomach can absorb are water, certain ions, drugs, and alcohol

- Questions ???